

Index

Page numbers in **bold** refer to main entries

A

abscesses, brain 197, **198**
 abstraction 151
 acetylcholine 24, 99
 action potential **22-23**, 136, 137
 action selection **33**
 actions
 conscious control over 168-169
 mirror neurons 102-103
 active efflux 19
 actual self 179
 adaptive behaviors 111
 adaptive forgetting 188
 ADD (attention deficit disorder) 216
 addiction 24, 25, 112, **113**, **212**
 adenine 56
 ADH (antidiuretic hormone) 95
 ADHD (attention deficit hyperactivity disorder) 47, 164, 186, **216**
 adipose tissue 94
 adoption 61
 adrenal gland 93
 adrenaline 93, 137, 208
 adrenocorticotrophic hormone (ACTH) 208
 adults
 brain **48-49**
 IQ 152
 male and female brains **58-59**
 perception of time 175
 personality 177
 aging
 aging brain **40-41**, 61
 how to slow the effects of **52-53**
 and time **175**
 aggression 38
 agnosia 39
 agonists 24
 alcohol 24, **25**, 52
 and language **125**
 and memory 146
 aldosterone 93
 alertness 28, 35
 alpha brain waves **131**
 alphabetic principle **131**
 altered states **170-171**
 alternating attention 165
 altruism **123**
 Alzheimer's disease 39, **50**, 51, 57, 185, 200
 amino acids 55
 amnesia 39, **146**, 197

amusia 215
 amygdala 32, 33, **38**, 39, 48, 59, 108, 213
 and emotions 107, 110, 111, 134
 and senses 77, 78
 analytical thinking 152
 anatomy **28-29**
 aneurysms 199
 anger 39, 106, **108-109**, 111, 116
 angiotensin 93, 95
 angular gyrus 127
 animals
 brains **14-15**
 hearing 77
 mirror neurons 102, 103
 with no brain 15
 sense of self 179
 vision 69
 anodal tDCS 191
 anorexia nervosa 214
 anoxia 206
 antagonists 24
 anterograde amnesia 146, 147
 antidepressants 111, 210
 antioxidants 55, 61
 anxiety 108, 109, 147, 185, 186
 disorders 47, 206, **208**, 213, 214
 apathy 200, 206
 aphasia **127**
 appetite 38
 arachnoid mater 16, 198
 arcuate fasciculus 151
 arteries 18, 19, 199
 articulation 126, 128, 131
 artificial intelligence 37, 163, **188-189**
 associations 65, 127, 142
 associative visual cortex 31
 astrocytes 17, 21, 173
 attachment 114, 115
 attention 24, 37, 51, 59, 100, **164-165**
 ADHD 47, 164, 186, **216**
 focusing **166-167**
 span 164, 207
 attraction **114**, 115
 auditory area 119, 183
 auditory canal 74
 auditory cortex 65, 74, 76, 127, 136
 auditory nerve 77
 augmented reality 162
 aura, and migraine 196
 autism spectrum disorders (ASD) 148, 215, **217**
 automatic functions 162
 autonomic nervous system 12, **13**, 92, 107
 autosomes 56, 57
 awareness 90, 96, 107, 162, 171, 173, 178

axons **20**, 21, 22-23, 26-27, 86, 98-99, 136

B

babies
 brain development **44-45**
 language learning **124-125**, 130
 sense of self 178, 179
 sight 70
 taste 81
 bacteria 198
 balance 50, 84, 98, 199
 basal ganglia **32-33**, 50, 51, 97, 174, 209
 basilar membrane 76
 belief **158-159**
 Bell's Palsy 203
 beta brain waves 42, 151, 163
 Big Five test 177
 bilingualism **125**
 binge-eating disorder (BED) 214
 binocular visual field 69
 bionic limbs **184**, 185
 bipolar disorder **207**
 bipolar neurons 20, 67
 bladder 13, 109
 blind spot **67**
 blind zone 69
 blood clots 199
 blood pressure 24, 36, 121, 199
 blood sugar levels 94, 109
 blood supply **18-19**, 109, 198, 199, 200
 blood vessels 17, 28, 61, 108
 blood-brain barrier **17**
 crossing **18-19**
 BMI 214
 body, brain in the 11, **12-13**
 body clock 207
 body language **118-119**, 121, 176, 177
 body shape 214
 bones
 cranium 16
 ear 74-75
 skeleton 98
 borderline personality disorder (BPD) 213
 brain cells *see* neurons
 brain damage
 before or during birth 204
 head injuries **197**
 and language **126**, **127**
 and learning 215
 and memory 139
 and morality 123
 and personality 177
 brain disorders 170, 174, 185, 190, **196-217**
 brain scans **40-41**

brain waves 42, 103, 151, 163, 168, 172
 brain/Cloud interface (B/CI) 192-193
 brain stem 28, 29, 32, 33, **36-37**, 44, 45, 200
 and emotion **186**
 and movement 97, 98
 and senses 74, 75, 81, 82, 84
 brain stem death **162**
 breathing 13, 90, 92, 108, 109, 162
 breathing exercises 88
 breathing rate 121
 Broca's area 31, 102, 119, 126, 127, 128, 131
 Brodmann areas 31
 Brodmann, Korbinian 31
 Bucy, Paul 39
 bulimia nervosa 214

C

caffeine 24, 174
 calcium/calcium ions 23, 26, 27
 cancer
 brain tumors **200**
 MRI scans 40
 capillaries 17, 18, 35
 carbohydrates 54
 carotid arteries 18, 19
 CAT (computerized axial tomography) scans 41
 cataplexy **205**
 catatonia 211
 cathodal tDCS 190
 caudate nucleus 32, 33, 121, 134
 cells
 brain **20-21**
 cell membranes 22-23
 cortex 31
 cellular wall 18
 central executive network 154, 155
 central nervous system (CNS) 12
 cerebellum 28, **36-37**, 44, 84, 97, 134, 200
 cerebral arteries 19
 cerebral cortex *see* cortex
 cerebral palsy 203, **204**, 215
 cerebral sinuses 18
 cerebrospinal fluid (CSF) **16-17**, 173, 198, 205
 cerebrum 29, 44, 200
 chemical weapons **23**
 chemicals, brain **24-25**
 chemoreceptors 82
 childbirth 34, 35, 206
 children
 brain development in older **46-47**
 brain development in young **44-45**
 core beliefs 159
 IQ 152